QUESTION NO:01

```
#include <iostream>
#include <string>
using namespace std;

class Vehicle {
private:
   string typeOfCar, make, model, color;
   int year;
   double milesDriven;

public:
   Vehicle(string t, string mk, string md, string c, int y, double m)
      : typeOfCar(t), make(mk), model(md), color(c), year(y), milesDriven(m) {}

   void displayVehicle() const {
      cout << "Type: " << typeOfCar << endl;
      cout << "Make: " << make << endl;
      cout << "Model: " << model << endl;
      cout << "Color: " << color << endl;
      cout << "Year: " << year << endl;
      cout << "Miles Driven: " << milesDriven << endl;
   }
```

```
};

class GasVehicle : virtual public Vehicle {
private:
   double fuelTankSize;

public:
   GasVehicle(string t, string mk, string md, string c, int y, double m, double fts)
      : Vehicle(t, mk, md, c, y, m), fuelTankSize(fts) {}

   void displayGas() const {
      cout << "Fuel Tank Size: " << fuelTankSize << endl;
   }
};

class ElectricVehicle : virtual public Vehicle {
private:
   double energyStorage;

public:
   ElectricVehicle(string t, string mk, string md, string c, int y, double m, double
es)
      : Vehicle(t, mk, md, c, y, m), energyStorage(es) {}
```

```
    void displayElectric() const {
        cout << "Energy Storage: " << energyStorage << endl;
    }
};

class HighPerformance : public GasVehicle {
private:
    int horsePower, topSpeed;

public:
    HighPerformance(string t, string mk, string md, string c, int y, double m,
double fts, int hp, int ts)
        : Vehicle(t, mk, md, c, y, m), GasVehicle(t, mk, md, c, y, m, fts),
horsePower(hp), topSpeed(ts) {}

    void displayHighPerformance() const {
        cout << "Horse Power: " << horsePower << endl;
        cout << "Top Speed: " << topSpeed << endl;
    }
};

class SportsCar : public HighPerformance {
public:
```

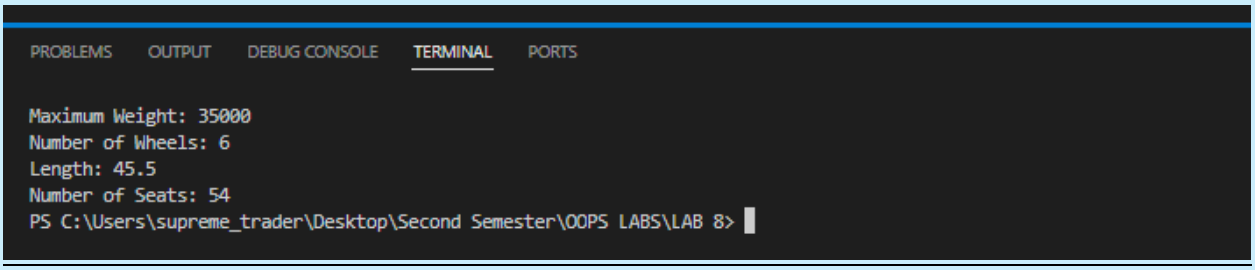
```
    string gearbox, driveSystem;

    SportsCar(string t, string mk, string md, string c, int y, double m, double fts, int hp, int ts, string gb, string ds)
        : Vehicle(t, mk, md, c, y, m), HighPerformance(t, mk, md, c, y, m, fts, hp, ts), gearbox(gb), driveSystem(ds) {}

    void displaySportsCar() const {
        displayHighPerformance();
        cout << "Gearbox: " << gearbox << endl;
        cout << "Drive System: " << driveSystem << endl;
    }
};

class HeavyVehicle : public GasVehicle, public ElectricVehicle {
private:
    double maximumWeight, length;
    int numberOfWheels;

public:
    HeavyVehicle(string t, string mk, string md, string c, int y, double m, double fts, double es, double mw, int nw, double len)
        : Vehicle(t, mk, md, c, y, m), GasVehicle(t, mk, md, c, y, m, fts), ElectricVehicle(t, mk, md, c, y, m, es),
```

```
        maximumWeight(mw), numberOfWheels(nw), length(len) {}

    void displayHeavy() const {
        cout << "Maximum Weight: " << maximumWeight << endl;
        cout << "Number of Wheels: " << numberOfWheels << endl;
        cout << "Length: " << length << endl;
    }
};

class ConstructionTruck : public HeavyVehicle {
public:
    string cargo;

    ConstructionTruck(string t, string mk, string md, string c, int y, double m,
double fts, double es, double mw, int nw, double len, string cg)
        : Vehicle(t, mk, md, c, y, m), HeavyVehicle(t, mk, md, c, y, m, fts, es, mw,
nw, len), cargo(cg) {}

    void displayConstructionTruck() const {
        displayHeavy();
        cout << "Cargo: " << cargo << endl;
    }
};
```

```
class Bus : public HeavyVehicle {
private:
   int numberOfSeats;

public:
   Bus(string t, string mk, string md, string c, int y, double m,
      double fts, double es, double mw, int nw, double len, int ns)
      : Vehicle(t, mk, md, c, y, m), HeavyVehicle(t, mk, md, c, y, m, fts, es, mw,
nw, len), numberOfSeats(ns) {}

   void display() const {
      displayVehicle();
      displayGas();
      displayElectric();
      displayHeavy();
      cout << "Number of Seats: " << numberOfSeats << endl;
   }
};

int main() {
   Bus myBus("Passenger Bus", "Volvo", "9700", "White",
         2023, 12500.5,
         150.0, 300.0,
```

```
        35000.0, 6, 45.5,
        54);

    myBus.display();

    return 0;
}
```

OUTPUT:



QUESTION NO:02

```
#include <iostream>
using namespace std;

class Character {
protected:
    string name;
    int level;
    int health;
```

```
public:
    Character(string n, int l, int h) {
        name = n;
        level = l;
        health = h;
    }

    void display() {
        cout << "Name: " << name << endl;
        cout << "Level: " << level << endl;
        cout << "Health: " << health << endl;
    }
};

class Warrior : virtual public Character {
protected:
    int strength;
    int meleeProficiency;

public:
    Warrior(string n, int l, int h, int s, int m)
        : Character(n, l, h) {
        strength = s;
        meleeProficiency = m;
    }
```

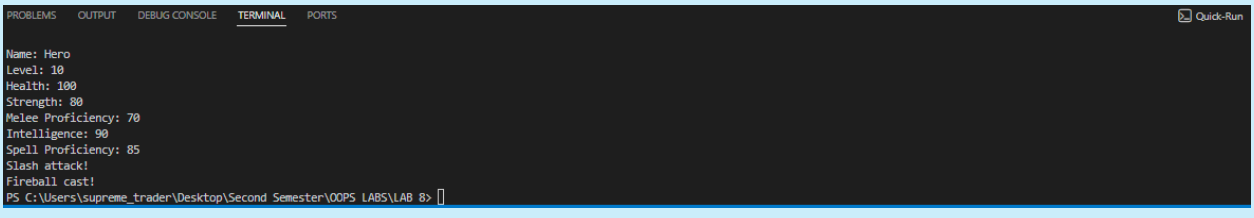
```
    void slash() {
        cout << "Slash attack!" << endl;
    }

    void display() {
        Character::display();
        cout << "Strength: " << strength << endl;
        cout << "Melee Proficiency: " << meleeProficiency << endl;
    }
};

class Mage : virtual public Character {
protected:
    int intelligence;
    int spellProficiency;

public:
    Mage(string n, int l, int h, int i, int sp)
        : Character(n, l, h) {
        intelligence = i;
        spellProficiency = sp;
    }

    void fireball() {
        cout << "Fireball cast!" << endl;
```

```
    }

    void display() {
        Character::display();
        cout << "Intelligence: " << intelligence << endl;
        cout << "Spell Proficiency: " << spellProficiency << endl;
    }
};

class Archer : public Character {
protected:
    int dexterity;
    int rangedProficiency;

public:
    Archer(string n, int l, int h, int d, int r)
        : Character(n, l, h) {
        dexterity = d;
        rangedProficiency = r;
    }

    void rapidShot() {
        cout << "Rapid shot!" << endl;
    }

    void display() {
```

```
        Character::display();
        cout << "Dexterity: " << dexterity << endl;
        cout << "Ranged Proficiency: " << rangedProficiency << endl;
    }
};

class NPC : public Character {
    string behavior;

public:
    NPC(string n, int l, int h, string b)
        : Character(n, l, h) {
        behavior = b;
    }

    void display() {
        Character::display();
        cout << "Behavior: " << behavior << endl;
    }
};

class Mighty : public Warrior, public Mage {
public:
    Mighty(string n, int l, int h, int s, int m, int i, int sp)
        : Character(n, l, h),
          Warrior(n, l, h, s, m),
```

```
    Mage(n, l, h, i, sp) {}

  void display() {
    Character::display();
    cout << "Strength: " << strength << endl;
    cout << "Melee Proficiency: " << meleeProficiency << endl;
    cout << "Intelligence: " << intelligence << endl;
    cout << "Spell Proficiency: " << spellProficiency << endl;
  }
};

int main() {
  Mighty m("Hero", 10, 100, 80, 70, 90, 85);
  m.display();
  m.slash();
  m.fireball();

  return 0;
}
```

OUTPUT:



QUESTION NO:03

```
#include <iostream>
using namespace std;

class Account {
protected:
   double balance;

public:

   Account() {
      cout << "Enter initial balance: ";
      cin >> balance;
   }



   Account(double b) {
      balance = b;
   }



   virtual void deposit(double amount) {
```

```
        balance += amount;
    }

    virtual void withdraw(double amount) {
        if (amount <= balance)
            balance -= amount;
        else
            cout << "Insufficient balance\n";
    }

    void checkBalance() const {
        cout << "Current Balance: Rs. " << balance << endl;
    }
};

class InterestAccount : virtual public Account {
protected:
    double interest;

public:
    InterestAccount() : Account() {
        char choice;
```

```
    cout << "Do you want to enter a custom interest rate? (y/n) [Default is 0.30]: ";
    cin >> choice;
    if (choice == 'y' || choice == 'Y') {
      cout << "Enter interest rate (e.g., 0.30 for 30%): ";
      cin >> interest;
    } else {
      interest = 0.30;
    }
  }

  InterestAccount(double b, double i = 0.30) : Account(b) {
    interest = i;
  }

  void deposit(double amount) override {
    balance += amount + (amount * interest);
    cout << "Deposited Rs. " << amount << " with interest applied.\n";
  }
};

class ChargingAccount : virtual public Account {
protected:
```

```
    double fee;


public:
    ChargingAccount() : Account() {
        char choice;
        cout << "Do you want to enter a custom withdrawal fee? (y/n) [Default is
Rs. 25]: ";
        cin >> choice;
        if (choice == 'y' || choice == 'Y') {
            cout << "Enter withdrawal fee: ";
            cin >> fee;
        } else {
            fee = 25.0;
        }
    }

    ChargingAccount(double b, double f = 25.0) : Account(b) {
        fee = f;
    }



    void withdraw(double amount) override {
        double total = amount + fee;
```

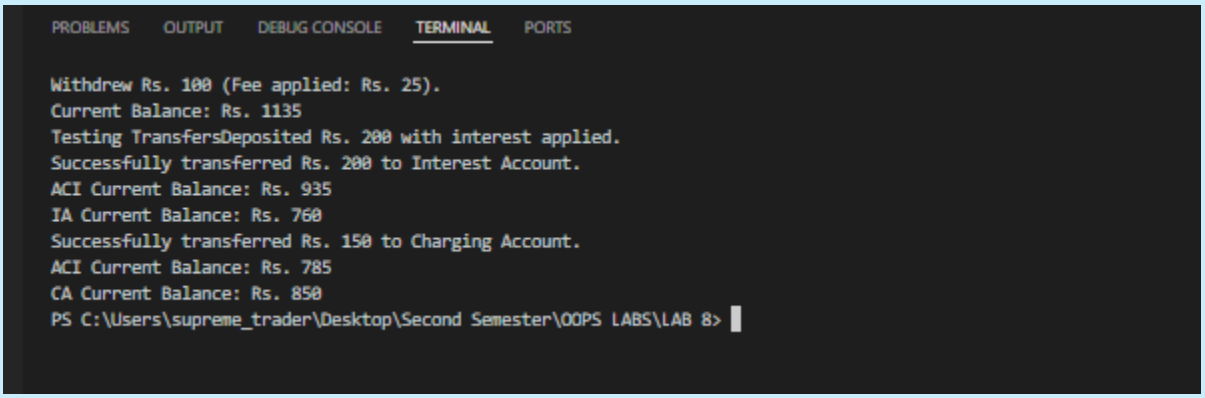
```
        if (total <= balance) {
            balance -= total;
            cout << "Withdrew Rs. " << amount << " (Fee applied: Rs. " << fee <<
").\n";
        } else {
            cout << "Insufficient balance to cover amount + fee.\n";
        }
    }
};

class ACI : public InterestAccount, public ChargingAccount {
public:

    ACI() : Account(), InterestAccount(), ChargingAccount() {}

    ACI(double b)
        : Account(b), InterestAccount(b), ChargingAccount(b) {}


    void transfer(double amount, Account &acc) {
        if (amount <= balance) {
            balance -= amount;
            acc.deposit(amount);
```

```
        cout << "Successfully transferred Rs. " << amount << " to Account.\n";
    } else {
        cout << "Transfer failed: Insufficient balance\n";
    }
}

void transfer(double amount, InterestAccount &acc) {
    if (amount <= balance) {
        balance -= amount;
        acc.deposit(amount);
        cout << "Successfully transferred Rs. " << amount << " to Interest
Account.\n";
    } else {
        cout << "Transfer failed: Insufficient balance\n";
    }
}

void transfer(double amount, ChargingAccount &acc) {
    if (amount <= balance) {
        balance -= amount;
        acc.deposit(amount);
        cout << "Successfully transferred Rs. " << amount << " to Charging
Account.\n";
    } else {
```

```
            cout << "Transfer failed: Insufficient balance\n";
        }
    }
};

int main() {
    cout << "--- Initializing Accounts ---\n";
    ACI a1(1000);
    InterestAccount ia(500);
    ChargingAccount ca(700);

    cout << "\n--- Testing ACI Functionality ---\n";
    a1.checkBalance();


    a1.deposit(200);
    a1.checkBalance();

    a1.withdraw(100);
    a1.checkBalance();

    cout << "Testing Transfers";
    a1.transfer(200, ia);
```

```
    cout << "ACI "; a1.checkBalance();
    cout << "IA "; ia.checkBalance();

    a1.transfer(150, ca);
    cout << "ACI "; a1.checkBalance();
    cout << "CA "; ca.checkBalance();

    return 0;
}
```

OUTPUT:



QUESTION NO:04

```
#include <iostream>
#include <string>
using namespace std;

class Media {
protected:
    string title;
```

```
    bool isBorrowed;

public:
    Media(string t) {
        title = t;
        isBorrowed = false;
    }

    void borrowMedia() {
        if (!isBorrowed) {
            isBorrowed = true;
            cout << title << " borrowed\n";
        } else {
            cout << title << " already borrowed\n";
        }
    }

    void returnMedia() {
        if (isBorrowed) {
            isBorrowed = false;
            cout << title << " returned\n";
        } else {
            cout << title << " was not borrowed\n";
        }
    }
```

```
    virtual void display() const {
        cout << "Title: " << title << endl;
        cout << "Status: " << (isBorrowed ? "Borrowed" : "Available") << endl;
    }
};

class BookDetails {
protected:
    string author;

public:
    BookDetails(string a) {
        author = a;
    }
};

class MagazineDetails {
protected:
    int issueNumber;

public:
    MagazineDetails(int i) {
        issueNumber = i;
    }
};
```

```
class DVDDetails {
protected:
   string director;

public:
   DVDDetails(string d) {
      director = d;
   }
};

class Book : public Media, public BookDetails {
public:
   Book(string t, string a) : Media(t), BookDetails(a) {}

   void display() const override {
      Media::display();
      cout << "Author: " << author << endl;
   }
};

class Magazine : public Media, public MagazineDetails {
public:
   Magazine(string t, int i) : Media(t), MagazineDetails(i) {}

   void display() const override {
      Media::display();
```

```
        cout << "Issue Number: " << issueNumber << endl;
    }
};

class DVD : public Media, public DVDDetails {
public:
    DVD(string t, string d) : Media(t), DVDDetails(d) {}

    void display() const override {
        Media::display();
        cout << "Director: " << director << endl;
    }
};

int main() {
    Book b("1984", "George Orwell");
    Magazine m("Time", 542);
    DVD d("Matrix", "Wachowski");

    b.display();
    cout << endl;

    b.borrowMedia();
    b.display();
    cout << endl;
```

```
    m.display();
    m.borrowMedia();
    m.returnMedia();
    cout << endl;

    d.display();

    return 0;
}
```

OUTPUT: